

DIPLOMA EXAMINATION IN  
ENGINEERING/TECHNOLOGY/MANAGEMENT/COMMERCIAL PRACTICE  
AUTOMOBILE CHASSIS

Time:3hours

(Maximum Marks; 100)

PART – A

(Maximum marks; 10)

I. Answer all questions in one or two sentences. Each question carries 2 marks

1. List any two objectives of suspension system.
2. Name two loads acting on front axle.
3. Define steering axis inclination.
4. What is fail safe brake.
5. List the uses of steering gear box.

(5X2=10)

PART-B

(Maximum marks; 30)

II. Answer any five of the following questions. Each question carries 6 marks

1. Describe the various types of sections used in frame construction.
2. Explain the working of double wishbone suspension.
3. Identify the functions of suspension system.
4. Explain under steer and over steer.
5. Sketch automobile chassis and name its components.
6. Compare disc brake and drum brake.
7. List the factors depend on stopping distance during braking.

(6X5=30)

PART-C

(Maximum marks; 60 )

Answer one full question from each unit. Each full question carries 15 marks

- III. (a). List and Explain different types of two wheeler frames. 8  
(b). Describe the method of checking the alignment of frame using plumb bob. 7

OR

- IV. (a). Sketch a front dead axle and mark the parts. 8  
(b). Sketch different arrangements of stub axle. 7

UNIT - 2

- V. (a). State the advantages of independent suspension system. 8  
(b). Write notes on leaf spring and helper spring. 7

OR

- VI. (a). Explain the working of telescopic shock absorber with the help of a figure. 8  
(b). Describe the working of air suspension. 7

UNIT - 3

- VII. (a). Write notes on wheel alignment. 8  
(b). Sketch and explain recirculating ball type steering gear box 7

OR

- VIII. (a). List the reasons for hard steering. 8  
(b). Explain the working of linkage type power steering. 7

UNIT - 4

- IX. (a). Explain the operation of drum brake with a help of a sketch. 8  
(b). Describe the working of wheel cylinder. 7

OR

- X. (a). Explain the working of disc brake. 8  
(b). List the advantages and disadvantages of hydraulic brake system. 7